GARGI GUPTA

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Education

PhD, SFI Center for Research & Training- Machine Learning Labs, Technological University Dublin, Ireland **S e p 2021- Ongoing**

Research area: Post hoc explanations for RNNs using state transition representations on univariate time series data (Explainable AI)

Master of Technology in Computer Science and Engineering, Jaypee Institute of Information Technology, Noida; CGPA: 9.0/10 **Jul 2018 – July 2020**

B. Tech Computer Science and Engineering, DIT University, Dehradun; GPA: 9.1/10.0 **Jul 2013 – May 2017**

Technical Skills

Languages and Tools: Python (Pandas, NumPy, SciPy Matplotlib, PyTorch), XAI, SHAP, C++, Orange, WEKA, AWS (EC2, IAM), Tableau, HTML, MS PowerPoint, ENVI 4.8, SNAP 5.0

Work Experience

- **Assistant Professor**, College of Engineering Roorkee (affiliated to Uttarakhand Technical University) **Aug 2020- Aug 2021**
  7th K.M. on Roorkee (NH-58), Rehmadpur Vardhmanpuram, Haridwar Rd, Roorkee, Uttarakhand 247667

  Teaching Subjects: Discrete Structures, Object Oriented Programming with JAVA & Principles of Programming Languages

- **Department Responsibilities**: OC Data Science Lab, Worked on NAAC Team (Criteria 7), Faculty Advisor, Project Coordinator (B. Tech First Year Students)

Internship Experience

Data Science Intern at Shutterstock, Dublin, Ireland **June 2023- Sep 2023**

- **Project Title**: **Explainability of the style encoding capabilities of the CLIP (Contrastive Language-Image Pre- Training) model for the AI Gen image**s. The analysis was carried out for AIGEN images with high-level, low-level, semantic and protected style. The dataset for these experiments was curated from the Shutterstock AIGEN image catalogue. Carried out an experiment and analysis on the style encoding capabilities of the CLIP model to identify, how effectively the CLIP model can differentiate between the AIGEN images and natural images.

Research Internship, Indian Institute of Technology Roorkee, Roorkee-247667 **Jan 2020 – Jul 2020**

- **Project Title: PolSAR Texture Feature Analysis for Generalized Land Cover Classification using Machine Learning & Deep Learning Techniques.** Worked on Sentinel-1A data. Used ENVI 4.8, SNAP 5.0 & MATLAB r2019 for data pre-processing and Analysis. Performed a detailed study and analysis of the role of texture features in Land Cover Classification.

Software Engineer Trainee, Infosys, No. 350, Electronics City, Hootagalli, Mysuru, Karnataka 570027 **Jan 2017 – May 2017**

- **Project Title**: **Mailing System (JAVA Based Enterprise Application)**. Built an intra-mailing system for the organization. The user could send emails with attachments and receive mail, along with various other features. Basic Knowledge and implementation of email protocols like SMTP, POP3, etc. Agile Methodology was used during development phase. Front-end of the application was developed using CSS, JavaScript, Bootstrap and AngularJS, HTML5. Framework used was Hibernate.

Summer Internship (Research based project), Indian Institute of Technology Roorkee, Roorkee-247667 **May 2015 – Jul 2015**

- **Project Title: Content based Image Retrieval: Using Color and Texture Feature.** Learnt about various image processing techniques and its application using MATLAB and proposed a new image retrieval techniques; local extrema co-occurrence patterns (LECoP) using the HSV color space.

Academic Projects

Processing and Analysis of Air Pollution Streaming Data Using Unsupervised Machine Learning Techniques

- Applied Data Stream clustering algorithms like k-means, BIRCH and DBSCAN. Applied incremental k-means clustering on new incoming air pollution streaming data to find the most dominant air pollutant in region under study. Dataset was created for Anand Vihar Location (Delhi, India). Visualizations and analysis was done using Tableau and Matplotlib.

Graph Based Recommendation System for Retail Stores

- Used Louvain Algorithm for community detection in retail store buyers' network and recommend the most suitable next product that can be brought by the buyer of the products from the retail store.

Publication & Media

- Jain, K., Gupta, G., Verma, K.K. *et al.* **Climatic Classification of India for Building Design Using Data Analytics**. *Natl. Acad. Sci. Lett.* **45**, 235–239 (2022). https://doi.org/10.1007/s40009-022-01109-7
- G. Gupta, M. Sharma, S. Choudhary and K. Pandey, "**Performance Analysis of Machine Learning Classification Algorithms for Breast Cancer Diagnosis**," *2021 9th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO)*, 2021, pp. 1-6, doi: 10.1109/ICRITO51393.2021.9596230.
- Gupta, Gargi, and Kumar Satyajeet. "**Impact analysis of Covid-19 lockdown on Delhi air quality: Using machine learning and data analysis techniques**."
- Gupta, G. and Kaur, P., **Processing and Analysis of Air Pollution Streaming Data: A Review** .
- "**A Machine Learning Approach for Digital Contact Tracing:TC4TL Challenge**", https://doi.org/10.48550/arXiv.2203.04307
- Doctoral Consortium Gupta, G., 2023. Post hoc explanations for RNNs using state transition representations for time series data.
- Gargi Gupta, Luca Longo, M. Atif Qureshi "**A Global Post Hoc XAI Method for Interpreting the LSTM using Deterministic Finite State Automata**," AICS Conference 2024, Accepted and Presented.
- ***"Explainable AI and the EU AI Act: Unlocking Trust and Compliance Before It's Too Late"***, Published by AI Ireland, April 2025.
- **Flash Talk**, *"Interpretable Time Series Classification using LSTM and Symbolic Abstraction"* Exploring Interdisciplinary Frontiers in Cognitive Science & AI, University of Cambridge –May 2025, Hosted by Cambridge C2D3, APSCI & Accelerate Programme.

Achievements

- Vice chancellor's gold medal for standing first in the branch of computer science and engineering of Master of Technology program of batch 2020 (Jaypee Institute of Information Technology, Noida, India)
- Fully funded PhD Scholarship by Science Foundation Ireland.
- Won 1st prize in PhD Bootcamp Team project "A Machine Learning approach to digital contact tracing: TC4TL Challenge"

Invited Talks/ Guest Speaker

- Invited as guest speaker at Department of Data Science, University of Galway (16 Nov 2022) at Galway Science and Tech Festival and Women in Data Science Seminar series. Video Link. Title of the Talk " Demystifying the Concept of Explainable AI".
- Guest Speaker at Pyladies Dublin, hosted by Digit Game Studio Dublin, Title of talk "Looking in depth overview of Explainable AI methods" (June, 2023).

Volunteer

- Community communications and support Women in AI, Ireland ,SFI center for Machine Learning Labs, Ireland Summer School Organizing Committee Member